MILITARY SPECIFICATION

CABLE, ELECTRICAL, SPECIAL PURPOSE, SHIP AND SHORE USE

1. SCOPE

1.1 Scope. This specification covers non-portable multiconductor cable for use in the AN/TPX-42A Interrogator System where the cable will be protected within racks, tunnels, buildings, ships, trailers or equipment.

1.2 Classification. Cables shall be of the following types and the configuration shown in Table I as specified in the contract or purchase order (see 6.1).

Type I - 44 Conductor cable
Type II - 13 Conductor cable
Type III - 54 Conductor cable
Type IV - 25 Conductor cable
Type V - 38 Conductor cable
Type VI - 49 Conductor cable

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-W-343 - Wire, Electrical (Uninsulated)

MILITARY

MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of
MIL-M-20693 - Molding Plastic, Polyamide (Nylon), Rigid

STANDARDS

FEDERAL

FED-STD-228 - Cable and Wire, Insulated; Methods Testing

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

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MIL-STD-681 - Identification Coding and Application of Hook Up and Lead Wire

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS
ASTM B33-63 - Tinned Soft or Annealed Copper Wire for Electrical Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.2 (see 6.2).

3.2 Material.

3.2.1 Conductors. The conductors shall be size number 20 American Wire Gage (AWG), composed of 19 strands of size number 32 AWG copper wire in accordance with Type C, Class C of QQ-W-343, tin coated, concentrically stranded.

3.2.1.1 Joining of conductors. Joints made in a conductor may be brazed using a silver alloy solder and a non-acid flux, or welded. Joints shall be butted and shall be free from lumps and sharp projections. The tensile strength of any section of a conductor joint which was brazed or welded shall be not less than 85 percent of the tensile strength of an adjacent section without a joint.

3.2.1.2 Conductor resistance. The Direct Current (DC) resistance of each conductor shall not exceed 13.12 ohms per 1,000 foot length when measured at, or corrected to, 25°C.

3.2.2 Insulation material. The conductor insulating material shall conform to type III, Grade E, of MIL-M-20693, 105°C.600VDC.

3.2.2.1 Insulation defects. The insulation shall be smooth and free from holes, splits, or blisters or other imperfections.

3.2.2.2 Tensile strength and elongation. The insulating material removed from the finished conductor shall meet the requirements specified herein, when tested in accordance with 4.7.

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Table I. Cable configuration.

Type Number	Number of Conductors	Internal construction			Overall construction		
		Component	Shield	Individual Conductor (Color Coding in accordance with MIL-STD-681)	Maximum Diameter	Shield	Sheath
II	13	5 Twist Pair	Yes	1, 8, 918, 928, 938, 948, 901	0.480	No	Yes
		1 Twist Triple	O/A	901, 908, 9017, 904, 913, 0			
IV	25	2 Twist Pairs	Yes	915, 916, 917, & 923	0.790	No	Yes
		1 Twist Triple	O/A	924, 925, & 926			
		1 Twist Quad	Yes	927, 934, 935, & 936			
		1 Twist Pair	O/A	937 & 945			
		6 Cond Insulated	Yes O/A	946, 947, 956, 957, 958, & 967			
		1 Twist Quad	Yes	968, 978, 9012, 9013,			
		1 Twist Pair	O/A	9014 & 9015			
V	38	2 Twist Pairs	Yes	915, 916, 917, 923,	0.995	No	Yes
		1 Twist Triple	O/A	924, 925, & 926			
		1 Twist Quad	Yes	927, 934, 935, 936			
		1 Twist Pair	O/A	937, 945			
		6 Cond Insulated	Yes O/A	946, 947, 956, 957, 956, 967			
		1 Twist Quad	Yes	968, 978, 9012, 9013, 9014, 9015			
		1 Twist Pair	O/A				
		5 Twist Pair	Yes	1, 8, 918, 928, 938, 948, 901			
		1 Twist Triple	O/A	908, 9016, 9017, 904, 913, 0			
I	44	20 Twisted Pairs	No	915, 916, 917, 923, 924, 925, 926, 927, 934, 935, 936, 937, 945, 946, 947, 956, 957, 958, 967, 968, 978,	0.810	Yes	Yes

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Table I. Cable configuration (continued).

Type Number	Number of Conductors	Internal construction			Overall construction		
		Component	Shield	Individual Conductor (Color Coding in accordance with MIL-STD-681)	Maximum Diameter	Shield	Sheath
				9012, 9013, 9014, 9015, 9016, 9017, 9018, 9023, 9024, 9025, 9026, 9027, 9028, 9034, 9035, 9036, 9037, 9038, & 9045			
		1 Twist Pair	Yes	9046 and 9047			
		1 Twist Pair	Yes	9048 and 9056			
VI	49	1 Twist Quad	Yes O/A	915, 916, 927, 923	0.955	No	Yes
		4 Cond Insulated	Yes				
		1 Twist Pair	O/A	924, 925, 926, 927, 934, 935			
		16 Cond Insulated	Yes	935, 937, 945, 946, 947, 956, 957, 958, 967, 968, 978, 9012, 9013, 9014, 9015, 9016, 0, 903,			
		2 Twist Pairs	O/A	9017, 9018			
		1 Twist Pair	Yes	9023, 9024, 9025, 9026, 9027, 9028			
		1 Twist Quad	O/A				
		1 Twist Quad	Yes O/A	9034, 9035, 9036, 9037			
		1 Twist Pair	Yes	9038, 9045, 9046, 9047, 9048			
		1 Twist Quad	O/A	9056			
		1 Twist Triple	Yes O/A	9057, 9058, 9067			
III	54	16 Cond Insulated	Yes O/A	915, 916, 917, 923, 924, 925 926, 927, 934, 935, 936, 937, 945, 946, 947, 956	0.990	No	Yes
		14 Cond Insulated	Yes O/A	957, 958, 967, 936, 978 9012, 9013, 9014, 9015, 9016, 9017, 9018, 9023, 9024			
		3 Twist Quad	Yes O/A	9025, 9026, 9027, 9028, 9034, 9035, 9036, 9037, 9038, 9045, 9046, 9047			
		2 Twist Six	Yes O/A	9048, 9056, 9057, 9058, 9067, 9068, 9078, 9123, 9124, 9125, 9126, 9127			

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Tensile strength, minimum pounds per square inch (psi). 1600
Elongation at rupture, minimum percent. 400

3.2.2.3 Color coding of insulation. The color coding for identification of each pair or group in the completed cable shall be as shown in Table I. The colored insulation on the individual conductors shall conform to the requirements of Table I, MIL-STD-681.

3.2.2.3.1 Insulation striping. All striping shall be readily discernable white stripes on the conductor insulation. These stripes shall be integral with the insulation. All striping shall remain intact through the life of the conductor insulation.

3.2.2.3.2 Stripe durability. A continuous line of the colored stripe shall not have been removed at the conclusion of the test specified in 4.6.

3.2.2.4 Twisting of pairs, triplets or quads. The insulated conductors shall be twisted into pairs, triplets or quads as specified in Table I. The length of lay of the twisted wires shall be designed to meet the capacitance unbalance requirements of 3.3.1. The average length of lay in any twisted group of wire shall not exceed 6 inches when measured on any 10 foot length.

3.2.3 Cable forming. The twisted groups shall be assembled to form a cylindrical core. Adjacent layers may be stranded in either the same direction or in opposite directions.

3.2.3.1 Tapes and binders. Tape wraps may be applied at the manufacturer's option. When tapes are used they shall be 0.001 inch thick polyethylene terepbthalate conforming to MIL-I-631, Type G or 0.005 inch to 0.010 inch polyvinyl chloride conforming to MIL-I-631, Type P, grade A. Tapes shall be applied with a minimum overlap of 25 percent.

3.2.4 Core covering. The core shall be covered with a sheath of mylar, polyethlene, or mylar backed "SBR" or equivalent, which shall be non-hygroscopic and shall be applied with an overlay.

3.2.4.1 Jacket over shield. A jacket over shielded conductors is not required provided the adjacent insulated conductors when in contact with the shielded conductors, are protected by mylar tape in accordance with MIL-I-631, Type G, one mil thickness with 50 percent overlay.

3.2.4.2 Four or more conductors. Four or more insulated conductors with an overall shield, shall be protected from the shield by mylar tape in accordance with MIL-I-631 Type G, one mil thickness with 50 percent overlay.

3.2.4.3 Dielectric. In each length of completed cable, the insulation between conductors and between conductors and shield shall withstand the dielectric DC potential for 3 seconds as follows:

	DC volts
Between conductors	3,600
Between conductors and shield	7,500

3.2.5 Shield. A shielding braid shall be woven of size number 34 AWG,

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Tin Coated copper strands conforming to ASTM-B33-63 before shielding. The shielding shall provide at least 85 percent coverage. The angle of the braid with the axis of the cable shall lie between 20 degrees or less for diameters up to 0.600 inch. For diamters larger than 0.600 inch the braid angle may be greater than 40 degrees. Percent coverage and angle of braid shall be calculated as follows:

$$K=(2F-F^2)X100$$

Where:

K = Percent coverage

a = Angle of braid with axis of cable

Tan a = $2\pi(D+2d)P/C$

d = Diameter of individual braid wire in inches

C = Number of carriers

D = Diameter of cable under shield in inches

F = NPd/sin a

N = Number of wires per carrier

P = Picks per inch of cable length

Unless otherwise specified, all shielding components shall be insulated by a component jacket to prevent electrical noise and stray ground currents.

3.2.5.1 Shield bending. The shield shall show no evidence of fracture or break visible ot the unaided eye after the shield bending test (see 4.4).

3.2.6 Jacket material. The cable jacket material shall be flame retardent polyurethane in accordance with MIL-I-631.

3.2.7 Sheath bending. The overall covering or sheath shall show no evidence of fracture or break visible to the unaided eye after the sheath bending test (see 4.5).

3.2.8 Cable marking. The cable shall be marked on the outside with the specification and type number and manufacturer's designation. A tape shall be permanently marked and inserted between the jacket and shield with the following information every 24 inches:

a. Manufacturer's name
b. Date of Manufacture
c. Contract Number
d. Specification Number

3.2.9 Cable construction. Cable construction shall be such as to provide good abrasion resistance, excellent flexibility, fluid resistance, under intended operation between -40°C and + 84°C.

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3.2.9.1 Cable shape. Cable shall be cylindrical and uniform with the maximum O.D. at any cross section as specified in Table I.

3.3 Electrical requirements.

3.3.1 Capacitance unbalance. The pair-to-pair and pair-to-shield unbalance tests shall be performed on the completed cable in accordance with 4.3.2 and 4.3.2.1.

3.3.2 Continuity. In each length of cable, the conductors shall be free from grounds (contacts between a conductor and shield) short circuits and open circuits. The shielding shall be free from open circuits.

3.3.3 Workmanship. Workmanship shall be such as to enable the cable to meet all applicable requirements of this specification, when inspected in accordance with section 4.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. First article samples shall be fabricated by the manufacturer with tools and methods that, as far as practicable are the same as those which will be used for quantity production of the product. The manufacture of the cable for a government contract, prior to approval of the first article sample, shall be at the manufacturer's risk. First article inspection shall be performed on a suitable length of cable. Certification of compliance for the basic material may be obtained from the supplier of the material providing that such certification contains actual test, examination or other verifiable quality data. First article inspection test reports shall be forwarded to the command or agency concerned, unless otherwise specified, for approval.

4.2.1 The first article samples shall be subjected to the tests listed in Table II. No failure shall be permitted.

Table II. First article inspection.

Examination and test	Requirement paragraph	Test paragraph
Visual and dimensional	3.2.1, 3.2.1.1, 3.2.2.3.1, 3.2.2.3.2, 3.2.8	4.3.1
Continuity	3.3.2	—
Conductor resistance	3.2.1.2, 3.2.2	4.9
Capacitance unbalance	3.3.1	4.3.2, 4.3.2.1

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Table II. First article inspection (Continued)

Examination and test	Requirement paragraph	Test paragraph
Insulation:		
Material	3.2.2	—
Defects	3.2.2.1	4.3.1
Tensile and elongation	3.2.2.2	4.7
Striping (if applicable)	3.2.2.3.1, 3.2.2.3.2	4.6
Color coding	3.2.2.3	4.3.1
Forming of pairs	3.2.2.4	—
Tapes and binders	3.2.3.1	—
Core covering	3.2.4, 3.2.6	—
Shield:		
Bending	3.2.5.1	4.4
Jacket:		
Material	3.2.6	—
Jacket over shield	3.2.4.1	—
Sheath bending	3.2.7	4.5
Dielectric	3.2.4.3	4.8

4.3 Test procedures.

4.3.1 Visual and dimensional examination. The color coding, shielding, core covering, marking tape and jacket thickness shall be examined at both ends of the shipping length on the reel. The cable shall be examined for jacket imperfections on all of the surface which is visible without unwinding the cable from the reel.

4.3.2 Capacitance unbalance. The capacitance unbalance test shall be performed on the cable shipping length in place on the reel. The pair-to-pair capacitance unbalance for 1,000-foot lengths of completed cable measured at 1,000 Hz at 10 percent shall not exceed the following:

Average for any reel	40 picofarads (pF)
Maximum individual for any reel	140 pF

For lengths of cable other than 1,000 foot, the value shall be converted to 1,000 foot values by dividing the unbalance of the length measured by the square root of the ratio of the length measured to 1,000. In cables with 50 pairs and less and in each group of multigroup cables the unbalances to be considered shall be:

(a) Between pairs adjacent in a layer.
(b) Between pairs in centers of four pairs or less.
(c) Between pairs in adjacent layers when the number of pairs in the inner (smaller) layer is six or less. In this, the center is counted as a layer.

The capacitance involved and test methods are as shown on figure 1. One and two represent the two conductors of a pair. Three and four represent the two conductors of a second pair. The capacitances, namely - C_{13}, C_{14}, C_{23}, C_{24},

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C_{12}, and C_{34} are the direct capacitances between conductors. The capacitances C_{1g}, C_{2g}, C_{3g}, and C_{4g} are the direct capacitances between wires 1 to 4, respectively, and all other conductors in the cable connected to the shield and grounded.

The pair-to-pair capacitance unbalance is:

$$(C_{14} + C_{23}) - (C_{13} + C_{24})$$

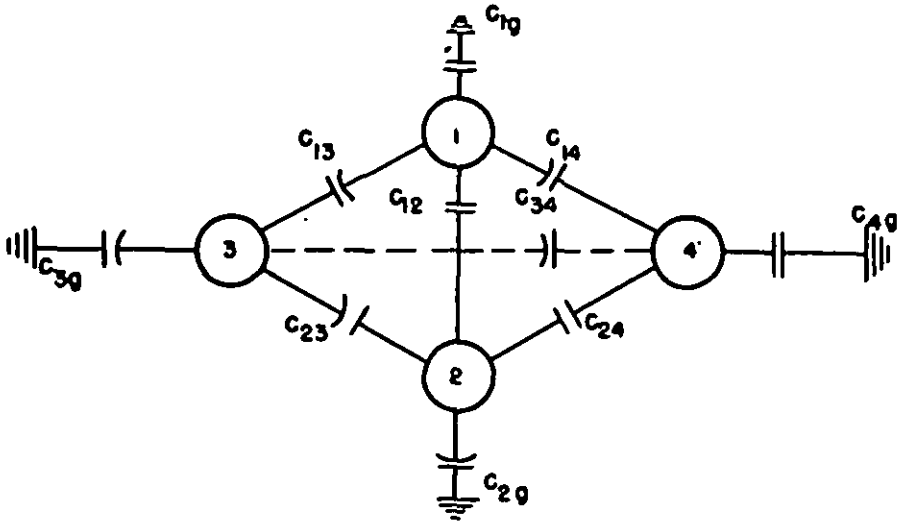


Figure 1. Test method for determining capacitance unbalance of pair to pair.

4.3.2.1 Capacitance unbalance, pair-to-shield. The pair-to-shield direct capacitance unbalance for 1,000-foot lengths of completed cable, when measured at 1,000 Hz ± 10 percent shall not exceed the average value for any reel of 100 pF. In figure 2, S indicates all wires (except conductors 1 and 2) bunched together. Wires 1 and 2 have a direct capacitance between them as well as a direct capacitance to the shield. Wires 1 and 2 also have direct capacitance to S and S has a direct capacitance to the shield.

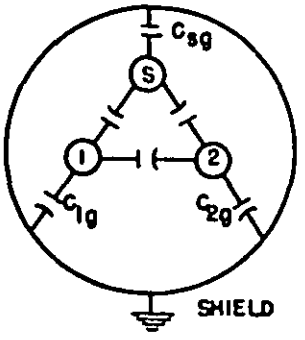


Figure 2. Test method for determining capacitance unbalance of pair-to-shield.

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4.4 Shield bending test. A 10 foot length of cable shall be bent at room temperature, with the shield overlay on the outside of the bend, in a 180° arc around a mandrel whose diameter is 20 times the outside diameter of the cable, straightened, then bent 180° in a reverse direction, completing one cycle. The sample shall then be straightened and rotated 90° about the longitudinal axis and a second cycle of bending performed. At the conclusion of this test the cable shield shall meet the requirements of 3.2.5.1.

4.5 Sheath cold bend test. Two specimens of cable shall be subjected to the cold bend test at the temperature indicated in Table III. The specimens shall be placed in a cold chamber in a non-flexed position and maintained at the required temperature for a minimum of 16 hours. Each specimen shall be bent around a mandrel with a diameter in accordance with Table III at the indicated speed for one complete turn prior to removal from the cold chamber. Upon removal from the chamber the specimen shall be examined and show no evidence of cracked sheath, component jacket or conductor insulation. The insulated conductors removed from the cable shall be submitted to the dielectric strength test of the applicable basic insulated wire specification.

4.6 Cable Stripe Test. A short specimen of finished wire or cable shall be firmly clamped in a horizontial position with its upper longitudial surface area freely exposed. A steel mandrel (0.025 diameter plus or minus 0.001 inch) shall be repeatedly rubbed over the surface at the stripe or mark so that the longitudinal axis of the mandrel and specimen shall be at right angles to each other. A weight shall so be attached to the jig holding the rubbing mandrel so that the combined jig and weight exerts a 500 gram thrust normal to the surface. A motor driven reciprocating cam mechanism and counter shall be used to permit an accurately measured number of abrasion strokes. The length of the stroke in one direction shall be 3/8 inch and the frequency of the stroke shall be 100 strokes per minute. The direction of the motion shall be along the axis of the specimen and perpendicular to the axis of the mandrel. The procedure shall be repeated on two additional specimens of wire selected 50 feet apart on a sample, to determine conformance to 3.2.2.3.2.

Table III. Cold bend test materials, speeds and temperatures

Cable size (outside diameter)	Mandrel Speed(RPM) / Temperature	1 / -40°C
0.000-0.300	MANDREL DIAMETERS	3XOD
0.301-0.350		3XOD
0.351-0.450		3XOD
0.451-0.550		4XOD
0.551-0.750		5XOD
0.751-0.850		6XOD
0.851-0.950		8XOD
0.951-1.500		10XOD

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4.7 Tensile and elongation test. Samples of the insulating material, shall be tested in accordance with FED-STD-228, Methods 3041 and 3031 respectively to determine conformance with 3.2.2.2.

4.8 Dielectric. The cable shall be subjected to a dielectric test to determine conformance with 3.2.4.3.

4.9 Resistance. The resistance of the conductors and shield shall be measured to determine conformance with 3.2.1.2 and 3.2.2.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation - packaging and packing. Preparation for delivery shall be in accordance with MIL-C-12000, at the level specified herein and in the contract or order.

5.1.1 Cables shall be furnished on non-returnable reels which contain a minimum length of 1000 feet or as specified in the contract. The cable lengths shall have both ends brought out so that cable can be readily tested without unreeling. The reels shall be constructed with flanges of sufficient thickness to protect the cable from damage during handling, shipment and storage. The diameter of the reel core shall be not less than 20 times the diameter of the cable.

5.2 Marking In addition to any special marking required by the contract or order, both flanges of each reel shall be marked in accordance with MIL-STD-129 and with the following information:

(a) Reel number
(b) Number of pairs in cable
(c) Length and location of each piece in feet for reels of multilengths
(d) Specific defective pair and cable length (if applicable)

5.2.1 Where practicable, metal tags, stencil, or paper labels containing the information specified in 5.2 shall be securely attached to the reels. Where paper labels are used, they shall be securely attached and protected by a transparent compound to prevent deterioration of the markings.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Cable type (see 1.2)
(c) Packaging, packing and marking requirements other than those required by 5.1 and 5.2

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples for those bidders offering a product which has been previously procured or tested by the Government and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement (see 4.2).

Preparing activity:
Navy - EC
(Project 6145-N237)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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